

Recursively defined sequences



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

Contents

1	Recursively defined sequences	2
2	Heron's algorithm	6
2.1	Error estimates for Heron's algorithm	11

1 Recursively defined sequences

- We begin with the following problem:

The bank lends us a_0 euros, with a management fee of b euros per year, at a fixed annual interest rate r ($r \in (0, 1)$). How much will we have to pay back after n years?

- After one year, the amount to be paid back will be

$$a_1 = a_0 + r a_0 + b = (1 + r) a_0 + b$$

after two years

$$a_2 = a_1 + r a_1 + b = (1 + r) a_1 + b$$

after three years

$$a_3 = a_2 + r a_2 + b = (1 + r) a_2 + b$$

and so on... Each year the amount is computed using the amount of the previous year. We therefore have the sequence:

$$\underbrace{a_0 \text{ given,}}_{\text{given value}} \quad \underbrace{a_{n+1} = (1 + r) a_n + b}_{\text{recursive relation}} \text{ with } n \in \mathbb{N}$$

This is a very simple case of a **recursively defined sequence**, in which each term is defined starting from the previous one (or the previous ones).

- In this case it is easy to convince ourselves that the sequence can be rewritten in **closed form** (i.e., explicitly).

After one year:

$$a_1 = (1 + r) a_0 + b$$

after two years:

$$\begin{aligned} a_2 &= (1 + r) a_1 + b = (1 + r) ((1 + r) a_0 + b) + b \\ &= (1 + r)^2 a_0 + b (1 + (1 + r)) \end{aligned}$$

after three years:

$$\begin{aligned} a_3 &= (1 + r) a_2 + b = (1 + r) \left((1 + r)^2 a_0 + b (1 + (1 + r)) \right) + b \\ &= (1 + r)^3 a_0 + b (1 + (1 + r) + (1 + r)^2) \end{aligned}$$

and so on ... hence

$$a_n = s^n a_0 + b(1 + s + s^2 + \cdots + s^{n-1}), \text{ where } s = (1 + r) > 1$$

We can therefore easily compute the limit as follows:

$$\lim_{n \rightarrow +\infty} a_n = \lim_{n \rightarrow +\infty} s^n a_0 + b(1 + s + s^2 + \cdots + s^{n-1}) = +\infty$$

However, it is not always possible to make explicit a sequence given in iterative form. In that case, we are left with the **problem of understanding its behavior and its limit**.

Example 1: Behavior of a recursively defined sequence

Given $b > 0$, we determine the behavior of the sequence defined by

$$a_0 = b, \quad a_{n+1} = \frac{a_n}{1 + a_n} \text{ with } n \in \mathbb{N}$$

All the terms of the sequence are positive, hence the denominator of the fraction appearing in the definition never vanishes.

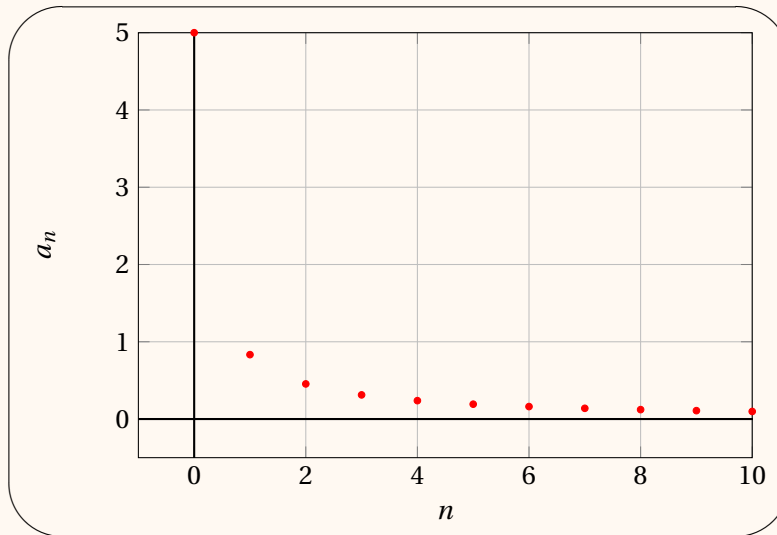
To verify this we proceed by induction:

$$a_0 = b > 0, \text{ and if } a_n > 0 \text{ then } a_{n+1} = \frac{a_n}{1 + a_n} > 0 \text{ (quotient of two positive numbers)}$$

Consequently, all the terms of the sequence after the initial one are less than 1, since $a_n < 1 + a_n$.

For example, with $b = 5$, we have:

$$a_0 = 5, \quad a_1 = \frac{5}{6}, \quad a_2 = \frac{\frac{5}{6}}{1 + \frac{5}{6}} = \frac{5}{11}, \quad a_3 = \frac{\frac{5}{11}}{1 + \frac{5}{11}} = \frac{5}{16} \dots$$



Example 2: Behavior of a recursively defined sequence

Let us try to understand whether or not the sequence is monotone. We know that $a_n > 0, \forall n \in \mathbb{N}$; then the inequality $a_n \leq a_{n+1}$ becomes

$$a_n \leq a_{n+1} \iff a_n \leq \frac{a_n}{1 + a_n} \iff 1 + a_n \leq 1 \iff a_n \leq 0$$

which we know to be false; hence the opposite inequality holds, i.e., the sequence is strictly decreasing ($a_n > a_{n+1}, \forall n \in \mathbb{N}$).

Since it is also bounded below (by zero), a_n converges to a finite non-negative limit, which we call $\ell \geq 0$.

We pass to the limit in the recursive relation.

$$\underbrace{\lim_{n \rightarrow +\infty} a_{n+1}}_{=\ell} = \lim_{n \rightarrow +\infty} \underbrace{\frac{a_n}{1 + a_n}}_{=\frac{\ell}{1+\ell}} \quad \text{hence} \quad \ell = \frac{\ell}{1+\ell} \quad \text{and consequently} \quad \ell = 0$$

since

$$\ell - \frac{\ell}{1+\ell} = 0, \quad \frac{\ell(1+\ell) - \ell}{1+\ell} = 0, \quad \frac{\ell^2}{1+\ell} = 0 \iff \ell = 0$$

Example 3: Closed formula of the recurrence

Given $b > 0$, we look for the closed formula of the sequence:

$$a_0 = b, \quad a_{n+1} = \frac{a_n}{1 + a_n} \quad \text{for } n \in \mathbb{N}$$

With $n = 1$, we have

$$a_1 = \frac{a_0}{1 + a_0}$$

With $n = 2$, we have

$$a_2 = \frac{a_1}{1 + a_1} = \frac{\frac{a_0}{1+a_0}}{1 + \frac{a_0}{1+a_0}} = \frac{\frac{a_0}{1+a_0}}{\frac{1+2a_0}{1+a_0}} = \frac{a_0}{1+a_0} \frac{1+a_0}{1+2a_0} = \frac{a_0}{1+2a_0}$$

With $n = 3$, we have

$$a_3 = \frac{a_2}{1 + a_2} = \frac{\frac{a_0}{1+2a_0}}{1 + \frac{a_0}{1+2a_0}} = \frac{\frac{a_0}{1+2a_0}}{\frac{1+3a_0}{1+2a_0}} = \frac{a_0}{1+2a_0} \frac{1+2a_0}{1+3a_0} = \frac{a_0}{1+3a_0}$$

and so on ... hence

$$a_n = \frac{a_0}{1 + n a_0} \quad \text{with } n \in \mathbb{N} \quad \text{and we have} \quad \lim_{n \rightarrow +\infty} \frac{a_0}{1 + n a_0} = 0$$

The previous example illustrates a good two-step strategy for studying the behavior of recursively defined sequences:

1. we try to understand whether the sequence satisfies some monotonicity property;
2. if so, we examine the possible limits, taking into account the recursive relation.

2 Heron's algorithm

- Given $b > 0$ and $c > 0$, we study the following recursively defined sequence:

$$a_0 = b, \quad a_{n+1} = \frac{1}{2} \left(a_n + \frac{c}{a_n} \right) \text{ with } n \in \mathbb{N}$$

It is evident that the sequence consists of positive numbers. To verify this we proceed by induction:

$$a_0 = b > 0, \text{ and if } a_n > 0 \text{ then } a_{n+1} = \frac{1}{2} \left(a_n + \frac{c}{a_n} \right) = \frac{1}{2} \left(\frac{a_n^2 + c}{a_n} \right) > 0$$

since we have the quotient of two positive numbers.

Let us try to understand whether or not the sequence is monotone. We know that $a_n > 0, \forall n \in \mathbb{N}$; the inequality $a_{n+1} < a_n$ becomes

$$a_{n+1} < a_n \iff \frac{1}{2} \left(a_n + \frac{c}{a_n} \right) < a_n \iff c < a_n^2 \iff a_n > \sqrt{c}$$

since

$$\frac{1}{2} \left(a_n + \frac{c}{a_n} \right) < a_n, \quad \frac{a_n^2 + c}{a_n} < 2 a_n, \quad a_n^2 + c < 2 a_n^2, \quad c < a_n^2$$

hence the sequence is strictly decreasing if and only if:

$$a_n > \sqrt{c}$$

We now need to understand whether or not this condition is true. We have

$$a_{n+1} = \frac{1}{2} \left(a_n + \frac{c}{a_n} \right) > \sqrt{c} \iff (a_n - \sqrt{c})^2 > 0 \iff a_n \neq \sqrt{c}$$

since

$$\frac{1}{2} \left(a_n + \frac{c}{a_n} \right) - \sqrt{c} > 0, \quad \frac{a_n^2 + c}{2 a_n} - \sqrt{c} > 0, \quad \frac{a_n^2 + c - 2 a_n \sqrt{c}}{2 a_n} > 0, \quad \frac{(a_n - \sqrt{c})^2}{2 a_n} > 0$$

hence $a_n > \sqrt{c}$ for every $n \geq 1$ if $\underbrace{b}_{=a_0} \neq \sqrt{c}$. And hence we always have $a_1 > \sqrt{c}, \forall b > 0$.

1. If $b > \sqrt{c}$ the sequence is strictly decreasing, therefore it has a limit (finite and positive), which we denote by $\ell \geq 0$.

To identify it, we pass to the limit in the recursive relation, obtaining

$$\underbrace{\lim_{n \rightarrow +\infty} a_{n+1}}_{=\ell} = \underbrace{\lim_{n \rightarrow +\infty} \frac{1}{2} \left(a_n + \frac{c}{a_n} \right)}_{=\frac{1}{2}(\ell + \frac{c}{\ell})} \text{ hence } \ell = \frac{1}{2} \left(\ell + \frac{c}{\ell} \right) \text{ and consequently } \ell = \sqrt{c}$$

since

$$\ell - \frac{1}{2} \left(\ell + \frac{c}{\ell} \right) = 0, \quad \ell - \frac{1}{2} \left(\frac{\ell^2 + c}{\ell} \right) = 0, \quad \left(\frac{2\ell^2 - \ell^2 - c}{2\ell} \right) = 0, \quad \left(\frac{\ell^2 - c}{2\ell} \right) = 0$$

Therefore, if $b > \sqrt{c}$, the whole sequence converges to $\sqrt{c}$ while decreasing.

2. If instead $0 < b < \sqrt{c}$, then by what we have seen before we will have $a_1 > \sqrt{c}$, and from that point on the sequence will start decreasing, again tending to $\sqrt{c}$.
3. Finally, if $b = \sqrt{c}$, the sequence remains constant, since

$$a_1 = \frac{1}{2} \left(\sqrt{c} + \frac{c}{\sqrt{c}} \right) = \frac{1}{2} \frac{\sqrt{c}\sqrt{c} + c}{\sqrt{c}} = \frac{1}{2} \frac{2c}{\sqrt{c}} = \frac{\sqrt{c}\sqrt{c}}{\sqrt{c}} = \sqrt{c}$$

and all the other terms a_n , with $n > 1$, will consequently have the same value $\sqrt{c}$.

- Moreover, $\forall b > 0$ and $n \geq 1$ we have:

$$\frac{c}{a_n} < \sqrt{c} \text{ since } a_n > \sqrt{c}, \quad c = \sqrt{c} \cdot \underbrace{\sqrt{c}}_{< a_n} \text{ and } c < \sqrt{c} \cdot a_n$$

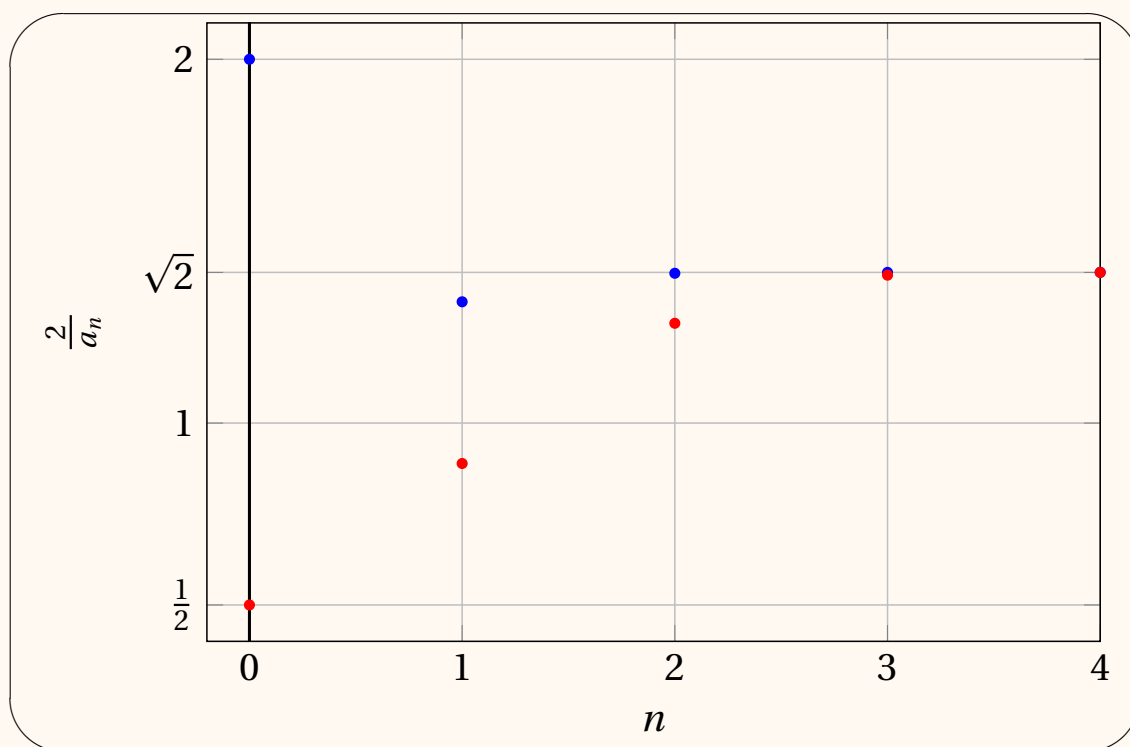
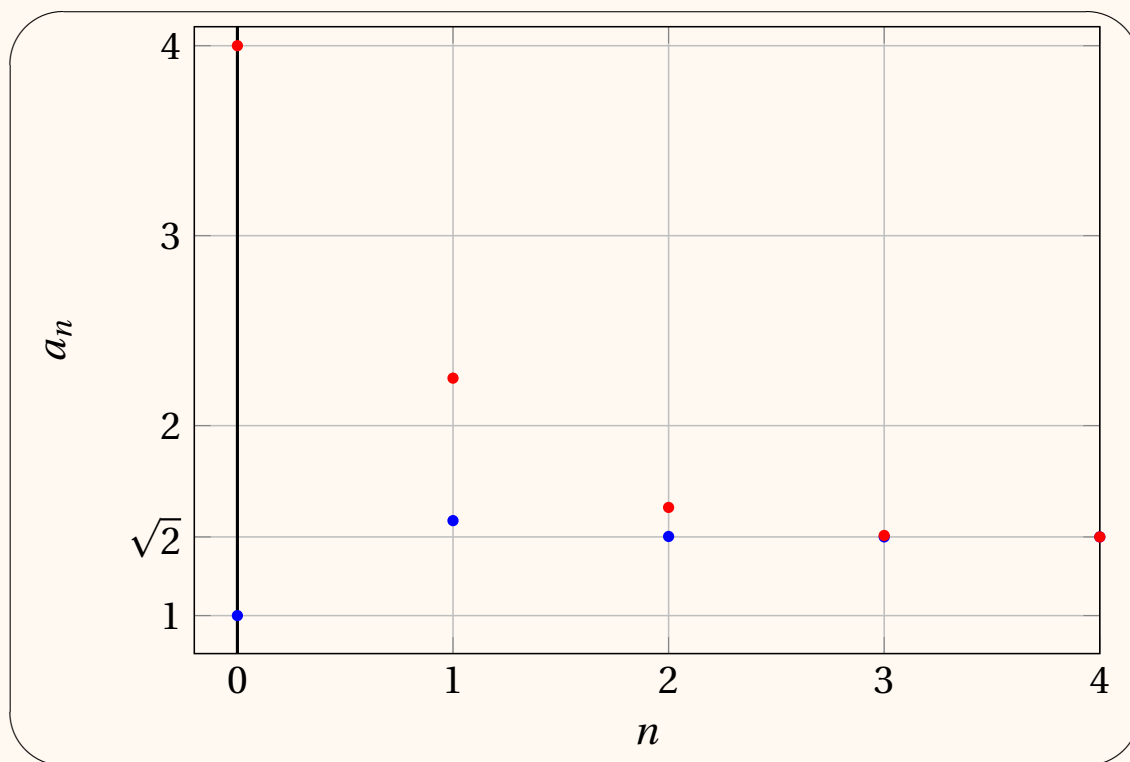
Summarizing, $\forall b > 0$ and $n \geq 1$ we have:

$$\frac{c}{a_n} < \sqrt{c} < a_n, \quad a_n \rightarrow \sqrt{c} \text{ and also } \frac{c}{a_n} \rightarrow \sqrt{c} \text{ since } c = \sqrt{c}\sqrt{c}$$

For every choice of $b > 0$ the sequence a_n tends to $\sqrt{c}$ and can therefore be used to approximate the square root.

Example 4: Behavior of the sequence

For example, with $c = 2$ and $b = 4 > \sqrt{2}$ (red points) or $b = 1 < \sqrt{2}$ (blue points), we have:



The recursive computation of the values of the sequence just seen is called Heron's **algorithm**, originally proposed to compute **the side of a square of area c** .

- The steps of the algorithm are:

1. Having fixed the base equal to a_0 (the initial value b), we construct the associated rectangle of area c . Its height will be $\frac{c}{a_0}$, since the area of the rectangle is fixed equal to that of the square, that is:

$$c = \left(a_0 \cdot \frac{c}{a_0} \right)$$

2. Since the side of the square we are looking for lies between a_0 and $\frac{c}{a_0}$, we take as the new approximation a_1 the average of a_0 and $\frac{c}{a_0}$:

$$a_1 = \frac{1}{2} \left(a_0 + \frac{c}{a_0} \right)$$

which becomes the new base of the rectangle. Then we take a_1 in place of a_0 and repeat, and so on.

For example, let us try to compute the side of a square of area $c = 2$; in other words, we try to estimate the value $\sqrt{2}$.

At the **first iteration** (with $n=0$) we consider, for example, a rectangle with base $a_0 = b = 4$ (the first upper **approximation** of $\sqrt{2}$); consequently, the height is $\frac{c}{a_0} = \frac{1}{2}$ (the first lower approx. of $\sqrt{2}$). Hence: $\frac{1}{2} < \sqrt{2} < 4$.

We compute the value a_1 :

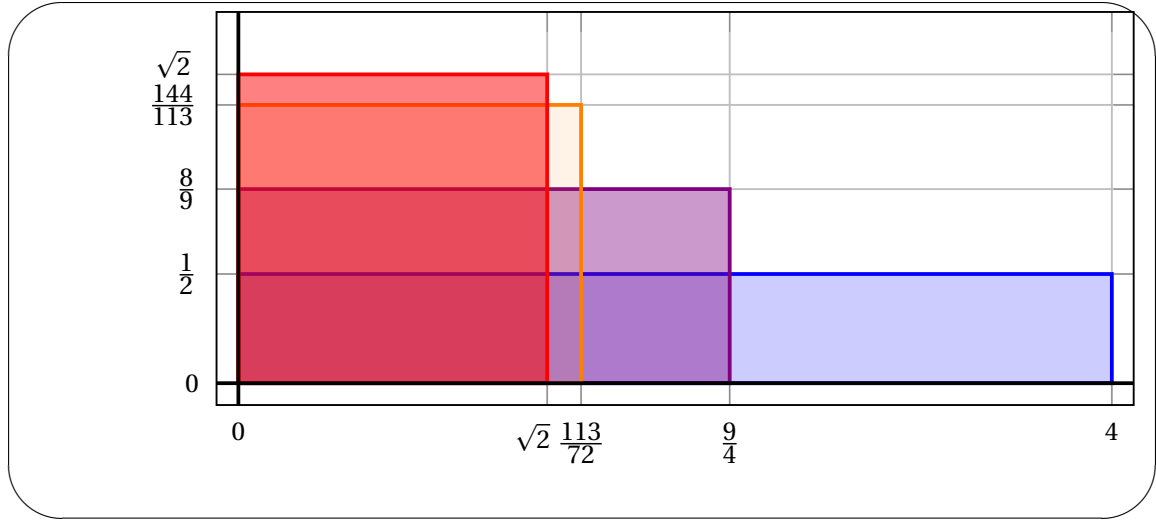
$$a_1 = \frac{1}{2} \left(4 + \frac{1}{2} \right) = \frac{9}{4}$$

At the **second iteration** (with $n=1$) we have a rectangle with base $a_1 = \frac{9}{4}$ (the second upper approx. of $\sqrt{2}$); consequently, the height is $\frac{c}{a_1} = \frac{2}{9/4} = \frac{8}{9}$ (the second lower approx. of $\sqrt{2}$). Hence: $\frac{8}{9} < \sqrt{2} < \frac{9}{4}$.

We compute the value a_2 :

$$a_2 = \frac{1}{2} \left(\frac{9}{4} + \frac{8}{9} \right) = \frac{113}{72}$$

At the **third iteration** (with $n=2$) we have a rectangle with base $a_2 = \frac{113}{72}$ (the third upper approx. of $\sqrt{2}$); consequently, the height is $\frac{c}{a_2} = \frac{2}{113/72} = \frac{144}{113}$ (the third lower approx. of $\sqrt{2}$). Hence: $\frac{144}{113} < \sqrt{2} < \frac{113}{72}$.



We compute the value a_3 :

$$a_3 = \frac{1}{2} \left(\frac{113}{72} + \frac{144}{113} \right) = \frac{23,137}{16,272}$$

At the **fourth iteration** (with $n=3$) we have a rectangle with base $a_3 = \frac{23,137}{16,272}$ (the fourth upper approx. of $\sqrt{2}$); consequently, the height is $\frac{c}{a_3} = \frac{2}{23,137/16,272} = \frac{32,544}{23,137}$ (the fourth lower approx. of $\sqrt{2}$). Hence: $\frac{32,544}{23,137} < \sqrt{2} < \frac{23,137}{16,272}$.

Already at the fourth iteration we have a fairly good estimate of $\sqrt{2}$:

$$\frac{23137}{16272} = 1.421890363... \quad \text{and} \quad \sqrt{2} = 1.414213562...$$

that is, an estimate that is correct up to the first decimal digit.

Finally, we compute the value a_4 :

$$a_4 = \frac{1}{2} \left(\frac{23,137}{16,272} + \frac{32,544}{23,137} \right) = \frac{1,064,876,737}{752,970,528}$$

With the new estimate we have:

$$\frac{1,064,876,737}{752,970,528} = 1.414234285... \quad \text{and} \quad \sqrt{2} = 1.414213562...$$

which is correct up to the fourth decimal digit, i.e., with just one additional iteration we have fixed 3 more decimal digits.

2.1 Error estimates for Heron's algorithm

- The **relative error** $\tilde{\varepsilon}_n$ at iteration n is:

$$\tilde{\varepsilon}_n = \frac{|a_n - \sqrt{c}|}{\sqrt{c}}$$

we have $\sqrt{c} < a_n, \forall b > 0$ and $n \geq 1$, hence we can simply consider:

$$\tilde{\varepsilon}_n = \frac{a_n - \sqrt{c}}{\sqrt{c}}$$

The value of $\sqrt{c}$ is unknown, hence we need a way to analyze the values of the relative errors that is independent of $\sqrt{c}$.

We first consider the **absolute error**:

$$\varepsilon_n = a_n - \sqrt{c}$$

we know that

$$\frac{c}{a_n} < \sqrt{c} < a_n, \quad \forall b > 0, n \geq 1$$

hence we can obtain an estimate of the absolute error ε_n made at iteration n as follows:

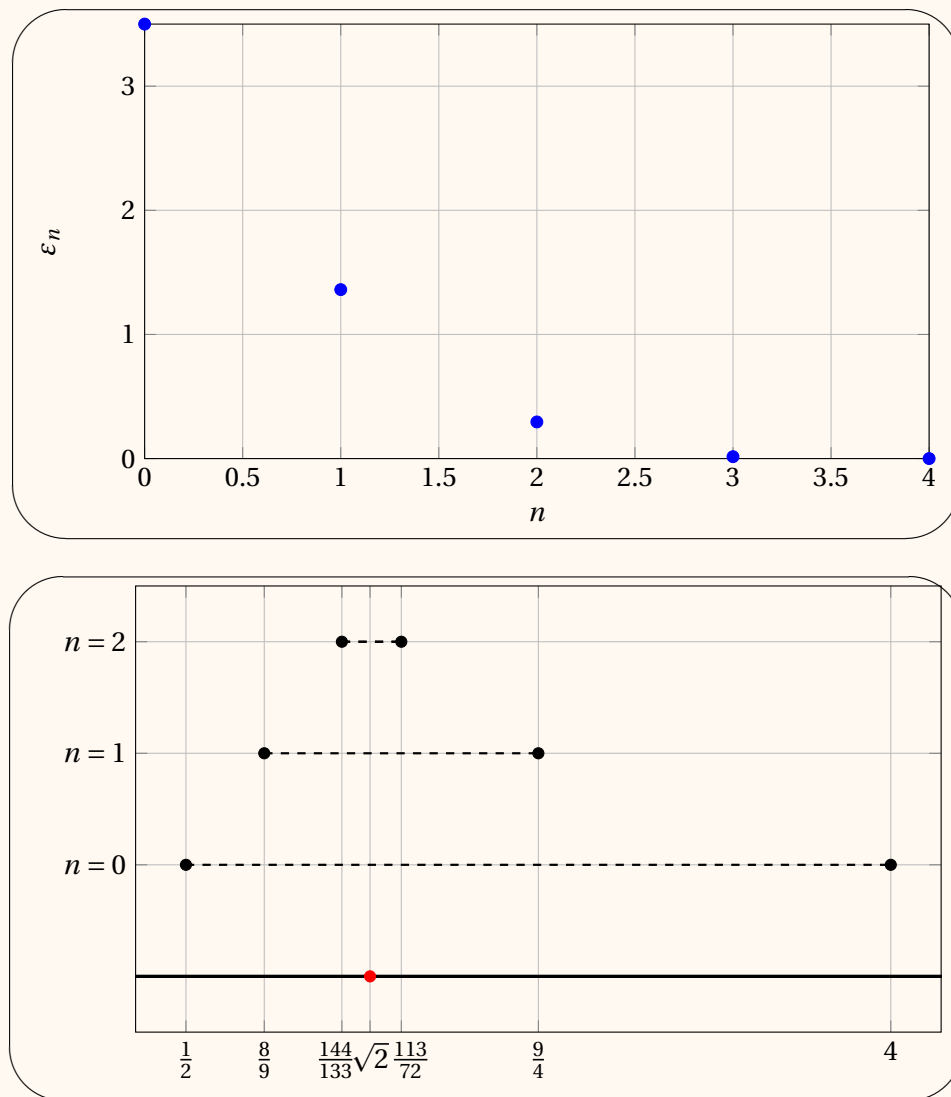
$$\varepsilon_n = a_n - \sqrt{c} < a_n - \frac{c}{a_n} \quad \text{and} \quad \varepsilon_n \rightarrow 0 \quad \text{for} \quad n \rightarrow +\infty \quad \text{since} \quad a_n \rightarrow \sqrt{c}$$

Example 5: Error estimates for Heron's algorithm ($\sqrt{2} = 1.414213562\dots$)

We go back to the previous example with $c = 2$ and $b = 4$, and we have:

iter.	$\frac{2}{a_n}$	a_n	estimate of $\sqrt{2}$	ε_n
$n = 0$	$\frac{1}{2}$	4	4	$< \frac{7}{2} = 3.5$
$n = 1$	$\frac{8}{9}$	$\frac{9}{4}$	2.25	$< \frac{49}{36} = 1.361\dots$
$n = 2$	$\frac{144}{113}$	$\frac{113}{72}$	1.569444444...	$< \frac{2,401}{8,136} = 0.295\dots$
$n = 3$	$\frac{32,544}{23,137}$	$\frac{23,137}{16,272}$	1.421890363...	$< \frac{5,764,801}{376,485,264} = 0.0153\dots$
$n = 4$	$\frac{1,505,941,056}{1,064,876,737}$	$\frac{1,064,876,737}{752,970,528}$	1.414234285...	$< \frac{33,232,930,569,601}{801,820,798,913,807,136} = 0.000041\dots$

Example 6: Errors and intervals considered by Heron's algorithm



We now want to understand how “fast” the error decreases by comparing the relative errors of two consecutive iterations. This estimate gives us information about the quality of the algorithm, i.e., we want to determine the “**rate of convergence**” of the algorithm.

We begin by observing that the value $\sqrt{c}$ can be written as follows:

$$\sqrt{c} = \frac{1}{2} \left(\sqrt{c} + \frac{c}{\sqrt{c}} \right) \text{ since } \frac{1}{2} \left(\sqrt{c} + \frac{c}{\sqrt{c}} \right) = \frac{1}{2} \frac{\sqrt{c}\sqrt{c} + c}{\sqrt{c}} = \frac{1}{2} \frac{2c}{\sqrt{c}} = \frac{\sqrt{c}\sqrt{c}}{\sqrt{c}} = \sqrt{c}$$

We now establish the relation between ε_{n+1} and ε_n ; we have:

$$\begin{aligned}
\varepsilon_{n+1} &= a_{n+1} - \sqrt{c} = \frac{1}{2} \left(a_n + \frac{c}{a_n} \right) - \frac{1}{2} \left(\sqrt{c} + \frac{c}{\sqrt{c}} \right) \\
&= \frac{1}{2} \left(a_n - \sqrt{c} + \frac{c}{a_n} - \frac{c}{\sqrt{c}} \right) = \frac{1}{2} \left(\varepsilon_n + \frac{c}{a_n} - \frac{c}{\sqrt{c}} \right) \\
&= \frac{1}{2} \left(\varepsilon_n + \frac{c(\sqrt{c} - a_n)}{\sqrt{c} a_n} \right) = \frac{1}{2} \left(\varepsilon_n - \frac{c \varepsilon_n}{\sqrt{c} a_n} \right) = \frac{1}{2} \left(\varepsilon_n - \frac{\sqrt{c} \varepsilon_n}{a_n} \right) \\
&= \frac{1}{2} \varepsilon_n \left(1 - \frac{\sqrt{c}}{a_n} \right) = \frac{1}{2} \varepsilon_n \left(\frac{a_n - \sqrt{c}}{a_n} \right) \\
&= \frac{1}{2} \frac{\varepsilon_n^2}{a_n}
\end{aligned}$$

We now look for the relation between $\tilde{\varepsilon}_{n+1}$ and $\tilde{\varepsilon}_n$, and we substitute $\varepsilon_n = \sqrt{c} \tilde{\varepsilon}_n$ into the previous formula:

$$\sqrt{c} \tilde{\varepsilon}_{n+1} = \frac{1}{2} \frac{(\sqrt{c} \tilde{\varepsilon}_n)^2}{a_n} = \frac{1}{2} \frac{\sqrt{c} \sqrt{c} \tilde{\varepsilon}_n^2}{a_n} \quad \text{hence} \quad \tilde{\varepsilon}_{n+1} = \frac{1}{2} \frac{\sqrt{c} \tilde{\varepsilon}_n^2}{a_n}$$

We have:

$$\tilde{\varepsilon}_n = \frac{\varepsilon_n}{\sqrt{c}} = \frac{a_n - \sqrt{c}}{\sqrt{c}} = \frac{a_n}{\sqrt{c}} - 1 \quad \text{hence} \quad \frac{\sqrt{c}}{a_n} = \frac{1}{\tilde{\varepsilon}_n + 1}$$

and hence we obtain:

$$\tilde{\varepsilon}_{n+1} = \frac{1}{2} \frac{\sqrt{c} \tilde{\varepsilon}_n^2}{a_n} = \frac{1}{2} \frac{\tilde{\varepsilon}_n^2}{1 + \tilde{\varepsilon}_n}$$

We have obtained the relation between $\tilde{\varepsilon}_{n+1}$ and $\tilde{\varepsilon}_n$, independent of both a_n and $\sqrt{c}$.

From the formula just obtained we can deduce the following simplified and more intuitive one:

$$0 \leq \tilde{\varepsilon}_{n+1} < \frac{1}{2} \min \{ \tilde{\varepsilon}_n, \tilde{\varepsilon}_n^2 \}$$

Hence:

$$\text{if } \tilde{\varepsilon}_n < 1, \text{ we have } \tilde{\varepsilon}_{n+1} < \frac{1}{2} \tilde{\varepsilon}_n^2$$

For example, if the relative error at a certain iteration equals 10^{-3} , at the next step it will be less than $5 \cdot 10^{-7}$. In other words, the number of zeros after the decimal point (which equals the number of correctly estimated digits after the decimal point) doubles at each iteration.

The relative error of Heron's algorithm at each step is proportional to the square of the relative error at the previous step. **The rate of convergence is quadratic.**